

Will Eaton

RESEARCH SUMMARY

My interests lie in the computational and theoretical aspects of seismology, in particular long-period phenomena such as Earth's normal modes, and the effects of motion on Earth's gravitational field.

EDUCATION

Princeton University, USA (2021 - 2026)

Ph.D. in Theoretical and Computational Seismology

Advisor: Professor Jeroen Tromp

University of Oxford, UK (2016 - 2021)

Integrated BA and MEarth Sci in Earth Sciences - First Class Honours

Advisor: Professor Tarje Nissen-Meyer

RESEARCH APPOINTMENT

Utrecht University, The Netherlands (2026 - present)

Assistant Professor

PEER-REVIEWED ARTICLES

In prep.	EATON, W. Computation of 2 nd spatial derivatives in the spectral-element method. <i>In preparation.</i> NISSEN-MEYER, T., LENG, K., FERNANDO, B., WOLF, J., HAINDL, C., EATON, W., ERMERT, L., MAG, A., RAMADAN, F., MUIR, J., THIYAGALINGAM, J., LONG., M. A versatile, efficient wave simulator for 1D, 2D, 3D structures across scales and complexities: AxiSEM3D. <i>In preparation.</i>
2025	EATON, W., GHARTI, H. N., TROMP, J. Sensitivity of prompt elasto-gravitational signals to 3D Earth structure, 2025. <i>Geophysical Journal International.</i>
2024	EATON, W., HAINDL, C., NISSEN-MEYER, T. Seismic scattering regimes from multiscale entropy and frequency correlations., 2024. <i>Geophysical Journal International.</i>
2023	GHARTI, H. N., EATON, W., TROMP, J. Spectral-infinite-element simulations of seismic wave propagation in self-gravitating, rotating 3D Earth models., 2023. <i>Geophysical Journal International.</i>

OTHER PUBLICATIONS

2024	FERNANDO, B., WOLF, J., LENG, K., NISSEN-MEYER, T., EATON, W., WALKER, A., CRAIG, T. J., MUIR, J., NUNN, C. AND LONG, M. AxiSEM3D - an introduction to using the code and its applications., 2024. <i>EarthArXiv.</i>
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INVITED SEMINARS

2024	Department of Earth, Environmental & Planetary Sciences, Brown University
2022	Jackson School of Geosciences, University of Texas at Austin

TEACHING EXPERIENCE

APC588: A Geometrical introduction to tensor calculus (Fall 2025)

Teaching assistant to Professor Jeroen Tromp - Princeton University

- Graduate-level class on differential geometry and tensor calculus.
- Designed, distributed, and graded weekly problem sets to students.

GEO441: Computational Seismology (Spring 2025)

Teaching assistant to Professor Jeroen Tromp - Princeton University

- Upper-level undergraduate course focused on strong and weak methods for solving hyperbolic and parabolic partial differential equations.
- Prepared the solution companion for reference textbook (theoreticalandcomputationalseismology.github.io) and presented weekly solutions on problems to class.

GEO103: Natural Disasters (Spring 2024)

Teaching assistant to Professor Blair Schoene - Princeton University

- Introductory course on natural hazards and plate tectonics designed for non-science majors.
- Prepared materials, conducted labs, graded, held office hours, and advised students on final projects.

AWARDS

2024	Outstanding Student Presentation Award	<i>American Geophysical Union</i>
2023	Myhrvold Havranek Family Venture Forward Fellowship	<i>Princeton University</i>
2021	Shell Prize	<i>University of Oxford</i>
	Schlumberger Prize	<i>University of Oxford</i>
2020	Gibbs Prize	<i>University of Oxford</i>
	Burdett-Coutts Prize	<i>University of Oxford</i>
2019	Keith Cox Prize	<i>University of Oxford</i>
	University College Scholarship	<i>University of Oxford</i>
2018	International Seismological Centre Prize	<i>University of Oxford</i>
2017	University College Exhibition	<i>University of Oxford</i>

CONFERENCE PROCEEDINGS

2025	EATON, W., BRETT, H., DEUSS, A., TROMP, J. , Probing tilted transverse inner-core anisotropy from normal-mode splitting. In <i>MODES 2025</i> (Oxford, UK, April 2025).
	GHARTI, H. N., EATON, W., J TROMP. , A frequency-domain spectral-element method for normal-mode simulations. In <i>MODES 2025</i> (Oxford, UK, April 2025).
2024	EATON, W., GHARTI, H. N., TROMP, J. , Sensitivity of prompt elasto-gravitational signals to 3D structure: implications for earthquake early-warning. In <i>AGU Fall Meeting 2024</i> (Washington DC, USA, December 2024).
	DEUSS, A., BRETT, H., EATON, W., J TROMP. , Tilted transverse isotropy in Earth's inner core from body waves and normal modes. In <i>AGU Fall Meeting 2024</i> (Washington DC, USA, December 2024).

2023	EATON, W., GHARTI, H. N., TROMP, J., Spectral-infinite-element modelling of GIA and sea-level change. In <i>POLENET 2023 GIA Training School</i> (Gävle, Sweden, July 2023).
	EATON, W., GHARTI, H. N., TROMP, J., Seismic wave propagation in self-gravitating Earth models with 3D heterogeneity. In <i>AGU Fall Meeting 2022</i> (Chicago, IL, December 2022).
2022	EATON, W., HAINDL, C., NISSEN-MEYER, T., The transition from ballistic to diffuse wavefields on Earth, its Moon and Mars. In <i>AGU Fall Meeting 2022</i> (Chicago, IL, December 2022). GHARTI, H. N., EATON, W., TROMP, J., Spectral-infinite-element simulations of seismic wave propagation in self-gravitating, 3D Earth models. In <i>SSA Seismic Tomography: What comes next?</i> (Toronto, Canada, October 2022).